

Lu Jiadong

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EDUCATION

National University of Singapore
Biomedical Informatics(Specialisation: Analytics)

Singapore
Aug 2025 - Now

Jilin University(Zhuhai) GPA:3.7/4.0
Data Science and Big Data Technology

Zhuhai, China
Oct 2020 - Jul 2024

WORK EXPERIENCE

AI Agent Engineer (7.2024 – 9.2025)

Jiusong Health Technology Co., Ltd.

- Developed a **multi-agent collaborative system** for personal diabetes care, abstracting complex care workflows into three specialized agents to enable evidence-based decision-making, long-term care interventions, and device-driven closed-loop management. (**paper under review**)
 - **Domain Expert Agent:** implemented a **RAG-based architecture** within Dify, orchestrating intent recognition, query rewriting, API retrieval, hybrid vector search, and memory management to improve the relevance and consistency of Care Guidance Information retrieval, supporting evidence-driven reasoning.
 - **Care Execution Agent:** Built a dynamic **care calendar** based on authoritative diabetes management SOPs to proactively trigger care tasks; developed the frontend config system supporting batch task operations, template imports, and scheduling to simplify care configuration and management (Tech stack: Vue3, TypeScript, SCSS).
 - **Data Analysis Agent:** Leveraged **MCP** to decouple LLM reasoning from deterministic numerical computation, enabling anomaly detection and event alerts on device time-series data; implemented polling-based real-time monitoring for blood glucose anomaly detection, and contributed to MCP server computation modules.

PROJECTS

- Collaborative LLM Interaction Policy Alignment (SFT + RLHF)
 - Constructed high-quality multi-turn collaborative instruction data (code & math) using GPT as a teacher model, and performed **full-parameter SFT on a 1.5B model**, focusing on task-completion-oriented interaction strategies to provide a stable policy initialization for subsequent RL training.
 - Designed a **multiturn-aware reward** using online dialogue simulation with an LLM-based user; introduced an exponential decay factor to penalize long trajectories, and incorporated **LLM-as-Judge (Gemini API)** to assess interaction quality, and applied Monte Carlo sampling to approximate expected rewards.
 - Conducted comparative RL training using **PPO / DPO / GRPO**, achieving an **18% accuracy improvement on reasoning tasks** over the base model and approximately **30% improvement in interaction-quality metrics**.
 - Tech Stack: HuggingFace Transformers & TRL, vLLM inference, Weights & Biases, FSDP-based distributed training on 2×A100 (40GB) GPUs.
- Large-Scale Exhibition Review Topic Modeling & Analysis (NLP / Unsupervised Learning)
 - Built an **unsupervised topic modeling pipeline** for exhibition reviews using embeddings → UMAP dimensionality reduction → HDBSCAN clustering, enabling automatic topic discovery while explicitly modeling noise samples.
 - To mitigate topic fragmentation, introduced **cluster-level similarity analysis based on cluster centroid embeddings**, identifying semantically overlapping clusters for merging.
 - Applied **LLMs (Gemini API)** in the post-clustering stage to generate unified topic labels and representative quotes, and produced topic-wise distribution statistics and visualizations to support reporting and operational issue analysis.

ADDITIONAL INFORMATION

- **Tools**, Git, Postman, Navicat, Docker, Jenkins, Overleaf, Trello, Miro, Termius
- **Cloud deployment**, Hands-on AWS experience, provisioning EC2/RDS, configuring VPC, and building CI/CD pipelines with Jenkins for front-end and back-end deployment.